MAHARASHTRA STATE BOARD OF TECHNICAL EDUCATION, MUMBAI

A Project Report on

Baby Vaccination Management System

Submitted as a partial fulfilment for the award of

Diploma in Computer Engineering

Submitted by

Salunkhe Ritika Sunil

Sonar Prerana Hari

Vyavahare Himanshu Ghanshyam

Wagh Kunal Sambhaji

DEPARTMENT OF COMPUTER ENGINEERING

S.S.V.P. S's Bapusaheb Shivajirao Deore Polytechnic, Dhule

(Institute Code: 0059)

2025 – 2026

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has satisfactorily completed project entitled **"Baby Vaccination Management System"** in partial fulfilment for the **Diploma Course in Computer Engineering** of the **Maharashtra State Board of Technical Education** at our polytechnic during the academic year **2025-2026**. The project is completed by a group consisting of four persons under the guidance of the Faculty Guide.

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Acknowledgment

We hereby take this opportunity to express our sincere gratitude towards all those who helped us in completing our project **"Baby Vaccination Management System"**.

While developing this project, we received valuable support and guidance from healthcare professionals and individuals who helped us understand the vaccination process. Their knowledge and inputs played an important role in the successful completion of this project.

We express our heartfelt thanks to our project guide Mr. N.S.Sharma for their continuous guidance, encouragement, and support throughout the project work.

We also extend our sincere thanks to HOD Mr. N.D.Patel & the teaching and non-teaching staff for their valuable suggestions and cooperation.

Finally, we would like to thank our parents and friends for their constant support and motivation.

Salunkhe Ritika Sunil

Sonar Prerana Hari

Vyavahare Himanshu Ghanshyam

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Index

Chapter Number	Title	Page Number
	Abstract	i
	List of Figures	ii
	List of Tables	iii
1	Introduction	1
2	Literature Survey 2.1 Study of Existing System 2.2 Limitations of Existing System 2.3 Problem Identification 2.4 Problem Definition	2
3	Scope of the Project 3.1 User Requirements 3.2 System Requirements	9
4	Methodology 4.1 Requirement Gathering and Analysis 4.2 Proposed Design	12
5	Details of Designs, Working and Processes	16
6	Results and Applications	23
7	Testing and Defect Report	36
8	Conclusion and Future Scope	43
9	References and Bibliography	44

Abstract

The proposed system **"Baby Vaccination Management System"** is developed to manage and track baby vaccination records in a digital format. This system helps parents and healthcare providers to monitor vaccination schedules and ensure timely vaccination of babies.

The application stores important details such as baby information, vaccination schedules, and vaccination history in a secure database. It allows parents to check upcoming vaccinations and helps doctors update vaccination records easily.

The system reduces manual paperwork, minimizes errors, and improves healthcare management. It provides a simple and efficient solution for maintaining vaccination data and ensuring better child healthcare.

List of figures

Fig No.	Title of Figure	Page No.
2.1	Manual Vaccination Record	3
2.3	Problem Identification Flow	5
5.1	System Workflow Diagram	16
5.2	System Architecture Diagram	18
5.3	Use Case Diagram	20
5.4	ER Diagram	22
6.1	Patient Dashboard Interface	24
6.2	Book Appointment Interface	25
6.3	Risk Status and Vaccination Monitoring Interface	26
6.4	Vaccination Card Interface	27
6.5	Vaccination Records Interface	28
6.6	Upcoming Vaccinations Interface	29
6.7	Vaccination Progress Interface	30
6.8	Doctor Dashboard Interface	31
6.9	Vaccine Records and Stock Management Interface	32
6.10	Staff Dashboard Interface	33
6.11	Appointment Management Interface	34

List of Table

Fig No.	Title of Table	Page No.
1	Test Cases	35
2	Defect Report	37

CHAPTER 1

INTRODUCTION

Healthcare plays a very important role in human life, especially in the early stages of a child's development. Vaccination is one of the most effective methods to protect babies from serious and life-threatening diseases. It helps in building immunity and ensures proper growth and development of children.

Vaccines protect babies from diseases such as polio, measles, hepatitis, tuberculosis, and many other infections. These diseases can be harmful and sometimes even fatal if proper vaccination is not given at the right time. Therefore, maintaining an accurate vaccination schedule is very important.

In many hospitals and healthcare centres, vaccination records are still maintained manually using paper-based systems. These manual systems are difficult to manage and often lead to problems such as loss of records, incorrect data entry, and difficulty in tracking vaccination schedules. Parents may forget vaccination dates due to lack of proper reminder systems.

India has a large population, and managing vaccination records for every child becomes a challenging task. Different vaccines are given at different stages of a child's growth, and it is important to follow the correct schedule. Missing a vaccine can affect the health of the child and increase the risk of diseases.

Parents often find it difficult to remember all vaccination dates and maintain records properly. At the same time, doctors and hospital staff face challenges in managing large amounts of patient data manually. This makes the system inefficient and time-consuming.

The Baby Vaccination Management System is developed to overcome these problems. It is a web-based application that helps parents, doctors, and hospital staff manage vaccination records digitally in an organized way.

The system stores important information such as baby details, vaccination schedules, and vaccination history in a secure database. It allows parents to check upcoming vaccination dates and helps doctors update vaccination status easily

CHAPTER 2

LITERATURE SURVEY

2.1 Study of Existing System

We studied the existing healthcare system related to baby vaccination management. In many hospitals and clinics, vaccination records are still maintained manually using paper-based registers.

We interacted with parents and healthcare staff to understand the current vaccination tracking system. During the discussion, the following questions were asked:

Q1. How are baby vaccination records maintained?
Q2. Do parents remember all vaccination dates?
Q3. How do doctors track vaccination schedules?
Q4. What problems occur in manual record keeping?
Q5. Is there any digital system used for vaccination tracking?
Q6. How is vaccination history maintained?
Q7. Do parents miss vaccination dates?

After discussing with them, we learnt that: -

Ans 1: Baby vaccination records are mostly maintained manually using paper registers or vaccination cards in hospitals and clinics.

Ans 2: No, parents do not always remember all vaccination dates, especially when there are many vaccines at different intervals.

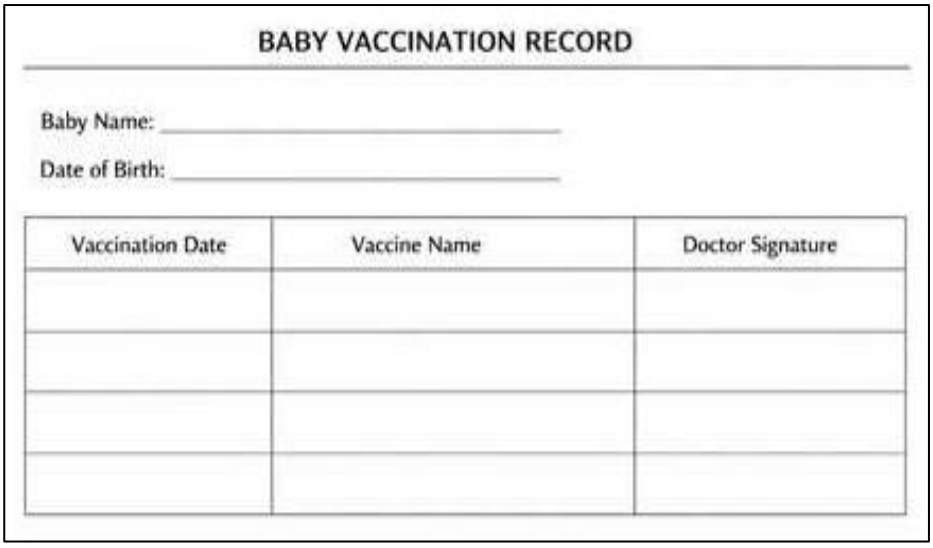
Ans 3: Doctors track vaccination schedules using manual records, registers, or vaccination charts.

Ans 4: Manual record keeping can cause problems such as data loss, errors, difficulty in tracking schedules, and time-consuming management.

Ans 5: Some hospitals use digital systems, but many small clinics still rely on manual methods.

Ans 6: Vaccination history is usually maintained in paper records or vaccination cards given to parents.

Ans 7: Yes, parents sometimes miss vaccination dates due to lack of reminders or proper tracking system



BABY VACCINATION RECORD

Baby Name: _____

Date of Birth: _____

Vaccination Date	Vaccine Name	Doctor Signature

Fig 2.1 Manual Vaccination Record

The above figure shows a manually maintained vaccination record used in hospitals or clinics. In this system, all the details of the baby such as name, date of birth, and parent information are recorded on paper. The vaccination date, vaccine name, and doctor's signature are also written manually.

This type of record keeping is simple but has several limitations. There are chances of data loss, damage, and difficulty in maintaining large records. It is also difficult to track vaccination schedules and history using manual records. Parents may forget vaccination dates as there is no reminder system available.

2.2 Limitations of Existing System

The existing system used for maintaining baby vaccination records is mostly manual and paper-based. Although this method has been used for many years, it has several limitations that affect the overall efficiency, accuracy, and reliability of healthcare management. As the number of patients increases, these limitations become more serious and create difficulties for both parents and healthcare providers.

One of the major problems with the existing system is that **records can be lost or damaged**. Since all the information is written on paper, there is always a risk of losing important documents due to misplacement, physical damage, or improper storage. Paper records can get torn, faded, or destroyed over time, which may result in the loss of important vaccination history of a baby.

Another important limitation is the **lack of an automatic reminder system**. In the manual system, there is no facility to notify parents about upcoming vaccination dates. As a result, parents may forget important vaccination schedules, which can affect the health of the child. Timely vaccination is very important, and missing even a single vaccine can lead to serious health risks.

Summary of Limitations

- Records can be lost, damaged, or misplaced
- No reminder system for vaccination dates
- Difficult to manage large data manually
- Time-consuming process
- High chances of human errors
- No centralized data storage
- Difficult to track vaccination history

2.3 Problem Identification

The study of the existing system shows that there are several problems faced by parents, doctors, and hospital staff while managing baby vaccination records. These problems arise mainly due to the use of manual systems.

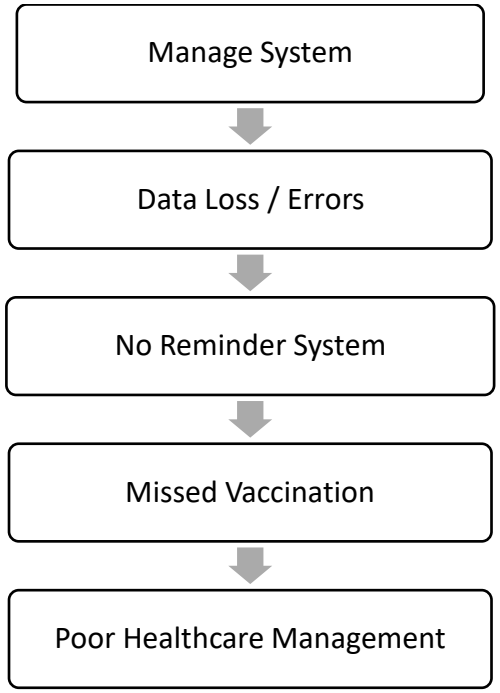


Fig 2.3 Problem Identification Flow

The above figure shows how problems arise in the existing manual system. Due to manual record keeping, data can be lost or contain errors. There is no reminder system, which leads to missed vaccinations. As a result, the overall healthcare management becomes inefficient and unreliable

Problems of Parent related issues

- Parents do not have a proper system to track upcoming vaccinations.
- Vaccination cards can be lost, damaged, or misplaced.
- Parents cannot easily access vaccination history when required.
- Lack of reminders leads to missed or delayed vaccinations.
- Parents face difficulty in understanding vaccination schedules properly.

Problems of Doctor related issues

- Doctors find it difficult to manage a large number of vaccination records manually.
- Tracking vaccination schedules for multiple babies becomes complex.

• Manual record keeping increases workload and reduces efficiency.
• Doctors cannot quickly retrieve past vaccination history of patients.
• Difficulty in maintaining accurate and updated records.

Problems of Hospital/Staff related issues

• Hospital staff face difficulty in maintaining and organizing records.
• Manual data entry is time-consuming and error-prone.
• Difficult to update and search records quickly.
• No proper reporting system is available.
• Managing data for multiple patients becomes complicated.

Problems of Existing System (Manual System)

• Records can be lost, damaged, or misplaced easily.
• No centralized database for storing information.
• No automatic reminder system for vaccination schedules.
• Difficult to manage large amounts of data manually.
• High chances of human errors in data entry.
• Difficult to track vaccination history and upcoming vaccines.

Overall Problems

• Lack of proper digital system for vaccination management
• Difficulty in tracking vaccination schedules
• Increased chances of missing vaccinations
• Inefficient healthcare management
• Time-consuming and unreliable system

2.4 Problem Definition

The existing system for maintaining baby vaccination records is mostly manual and paper-based, which creates several challenges for parents, doctors, and healthcare staff. Due to these limitations, there is a need to develop an efficient and reliable system that can manage vaccination records in a better way.

The main objective of this project is to develop a **web-based Baby Vaccination Management System** that will facilitate easy and effective record keeping of vaccination details.

The proposed system aims to solve the problems faced in the existing system by providing a digital platform for managing vaccination data.

The system will focus on:

a) Maintaining proper records related to baby vaccination details

b) Managing vaccination schedules and history in an organized manner

c) Providing easy access to data for parents and doctors

To achieve this, the proposed system will provide:

a) **Detailed categorization of vaccination records:**
The system will store all vaccination data in a structured format, including baby details, vaccine name, date, and doctor information. This will help in organizing data properly.

b) **Provision to maintain vaccination details:**
The system will allow users to add, update, and view vaccination records easily. This will reduce manual work and improve data accuracy.

c) **Reminder and notification system:**
The system will provide reminders for upcoming vaccination dates so that parents do not miss important vaccinations.

d) **Centralized database management:**
All records will be stored in a centralized database, making it easy to access data anytime and from anywhere.

e) **Generation of reports:**

The system will generate reports of vaccination history, which will help doctors and parents to track progress effectively.

Benefits of Proposed System

- Reduces chances of data loss
- Improves accuracy of records
- Saves time for doctors and staff
- Helps parents remember vaccination schedules
- Provides better healthcare management

CHAPTER 3

SCOPE OF THE PROJECT

3.1 Scope of the Project

The proposed system, **Baby Vaccination Management System**, is designed to simplify and improve the process of managing vaccination records for babies. In the existing system, vaccination records are maintained manually, which leads to various problems such as data loss, difficulty in tracking, and lack of reminders. The scope of this project focuses on overcoming these limitations by providing a digital and efficient solution.

The application aims to provide a centralized platform where all vaccination-related information can be stored, managed, and accessed easily. This system will help both parents and healthcare providers in maintaining accurate and up-to-date vaccination records.

The scope of the project includes managing all aspects of vaccination, starting from baby registration to tracking vaccination schedules and maintaining complete vaccination history. The system ensures that all data is stored securely and can be accessed whenever required.

Key Functional Areas of the System

The scope of this project is divided into several functional modules which help in efficient management of the system.

1) Baby Registration and Management

- The system allows users to register baby details such as name, date of birth, parent details, and contact information.
- Each baby will have a unique record in the system.
- This helps in maintaining organized and structured data.
- The system allows updating and viewing baby details anytime.

2) Vaccination Schedule Management

- The system automatically generates vaccination schedules based on the baby's age.

• It helps parents to know upcoming vaccination dates.
• The system ensures that all vaccinations are scheduled properly according to medical guidelines.
• It reduces the chances of missing any vaccination.

3) Vaccination Record Management

• The system stores detailed vaccination records such as vaccine name, date, and doctor details.
• It allows adding, updating, and viewing vaccination records.
• All records are stored in a digital format, reducing dependency on paper.
• It helps in maintaining complete vaccination history.

4) Doctor and Admin Management

• Doctors can update vaccination details after each visit.
• Admin can manage the entire system including users and data.
• The system allows easy access to patient records for healthcare providers.
• It reduces workload and improves efficiency.

5) Data Storage and Security

• All data is stored in a centralized database.
• It ensures data safety and reduces chances of data loss.
• Secure login system protects user information.
• Data can be accessed from anywhere when required.

6) Report Generation

• The system generates detailed reports of vaccination history.
• Reports help in tracking completed and pending vaccinations.
• Doctors and parents can easily analyse the data.
• It supports better decision-making in healthcare.

Benefits of the Proposed System

- Reduces manual work and paperwork
- Improves accuracy of records
- Provides timely reminders
- Ensures proper tracking of vaccination schedules
- Saves time for doctors and parents
- Enhances overall healthcare management

Future Scope of the Project

The system can be further enhanced with additional features such as:

- Mobile application for easy access
- SMS and email notification system
- Multi-language support for different regions
- Integration with government healthcare systems
- Cloud-based storage for better accessibility

CHAPTER 4

METHODOLOGHY

4.1 Requirements Gathering and Analysis

For the development of the **Baby Vaccination Management System**, The linear and sequential process where each phase is completed before moving to the next phase. This model is suitable for this project because the requirements are clearly defined and structured.

The Software Development Life Cycle (SDLC) followed in this project includes the following phases:

1. Requirements Gathering and Analysis
2. System Design
3. Implementation
4. Testing
5. Deployment
6. Maintenance

Phase 1: Requirements Gathering and Analysis

In this phase, we focused on understanding the needs of parents, doctors, and healthcare staff. The main objective was to identify the problems in the existing system and define the requirements of the proposed system.

We collected information through:

- Discussions with parents
- Interaction with doctors
- Observation of existing manual systems

This helped us to design a system that solves real-world problems effectively.

4.1.1 User (Parent) Requirements

The system should fulfil the following requirements for parents:

- The system should allow parents to register baby details.
- It should store information such as baby name, date of birth, and parent contact details.
- Parents should be able to view vaccination schedules easily.
- The system should provide reminders for upcoming vaccinations.
- Parents should be able to access vaccination history anytime.
- The system should be simple and user-friendly.

4.1.2 Doctor Requirements

- The system should allow doctors to update vaccination details.
- Doctors should be able to view patient records quickly.
- The system should reduce manual work.
- Doctors should be able to track vaccination history of babies.
- It should improve efficiency in managing patient data.

4.1.3 Baby (Record) Requirements

- The system should accept and store baby details.
- Each baby should have a unique record in the system.
- The system should maintain vaccination details for each baby.
- It should store data securely in the database.

4.1.4 Vaccination Requirements

- The system should maintain a list of vaccines.
- It should generate vaccination schedules automatically.
- It should store vaccine name, date, and status.
- It should track completed and pending vaccinations.
- It should provide reminders for upcoming vaccines.

4.1.5 Record Management Requirements

- The system should store all data in a structured format.
- It should allow adding, updating, and deleting records.

- It should maintain vaccination history.
- It should provide easy access to stored data.

4.1.6 System Requirements

Software Requirements

Operating System: Windows OS

Platform: Visual Studio Code

Languages: HTML, CSS, JavaScript, Node.js

Database: MongoDB

Hardware Requirements

Processor: Intel i3 or higher

RAM: Minimum 4 GB

Hard Disk: 500 GB or more

Phase 2: System Design

In this phase, the system architecture, database structure, and user interface were designed.

- Database design includes tables for users, babies, and vaccination records
- User interface is designed for easy navigation
- System modules are defined clearly

Phase 3: Implementation

In this phase, the system is developed using:

- Frontend: HTML, CSS, JavaScript, React
- Backend: Node.js
- Database: MongoDB

All modules such as registration, vaccination tracking, and reporting are implemented.

Phase 4: Testing

Testing is done to ensure the system works correctly.

- Unit Testing – testing individual modules
- Integration Testing – testing combined modules
- System Testing – testing the complete system

Errors and bugs are identified and fixed.

Phase 5: Deployment

- The system is deployed on a server
- Users can access the system through a web browser
- System becomes ready for real-time use

Phase 6: Maintenance

- Regular updates are performed
- Bugs are fixed if found
- System performance is improved over time

CHAPTER 5

DETAILS OF DESIGN, WORKING AND PROCESSES

System Design

In the System Design phase of the **Baby Vaccination Management System**, the main objective was to convert the gathered requirements into a structured and well-organized system. This phase plays a crucial role in defining how the system will function and how different components will interact with each other.

The system design includes:

- System Architecture
- Database Design
- Module Design
- User Interface Design

The design ensures that the system is efficient, user-friendly, and capable of handling all required operations smoothly.

Fig 5.1: System Workflow Diagram

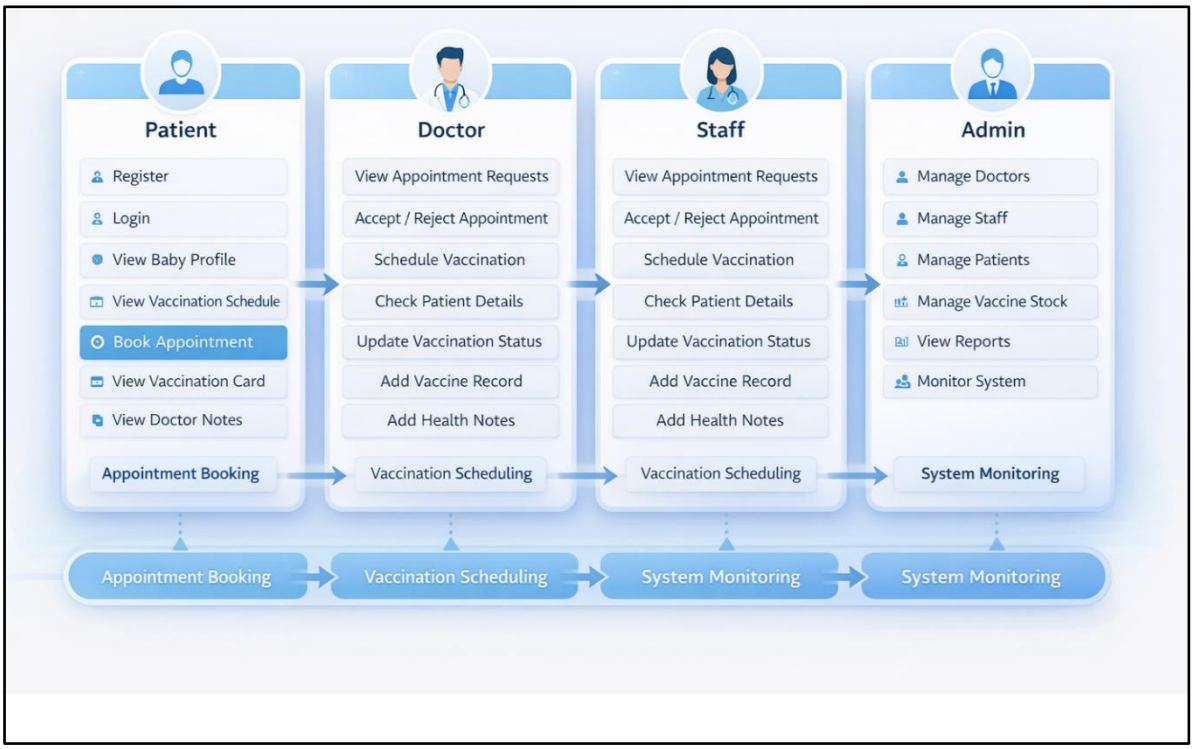
The above figure represents the overall working of the Baby Vaccination Management System, which involves four main users: Patient, Doctor, Staff, and Admin. The system is designed to ensure smooth communication and efficient management of vaccination processes.

Initially, the process starts with the patient. The patient registers into the system, logs in, and views the baby profile along with the vaccination schedule. Based on the schedule, the patient can book an appointment for vaccination.

Once the appointment is booked, the request is received by the doctor and staff. They have the authority to accept or reject the appointment. After accepting, they schedule the vaccination and verify patient details. Following the vaccination, they update the vaccination status, add vaccine records, and provide necessary health notes.

After the process is completed, the patient can view the vaccination card, check vaccine status, receive notifications, and also read doctor's notes for further guidance.

The admin plays a vital role in managing the entire system. The admin is responsible for managing doctors, staff, patients, and vaccine stock. Additionally, the admin monitors the system activities and generates reports to ensure smooth functioning.



Step-wise Working of System

1. Patient registers and logs into the system
2. Patient views baby profile and vaccination schedule
3. Patient books appointment for vaccination
4. Doctor/Staff receive appointment request
5. Appointment is accepted or rejected
6. Vaccination is scheduled
7. Patient details are verified
8. Vaccination is performed
9. Vaccination status is updated
10. Vaccine record is added
11. Health notes are provided

12. Patient views vaccination card and status

13. Notifications are received by patient

14. Admin manages entire system and generates reports

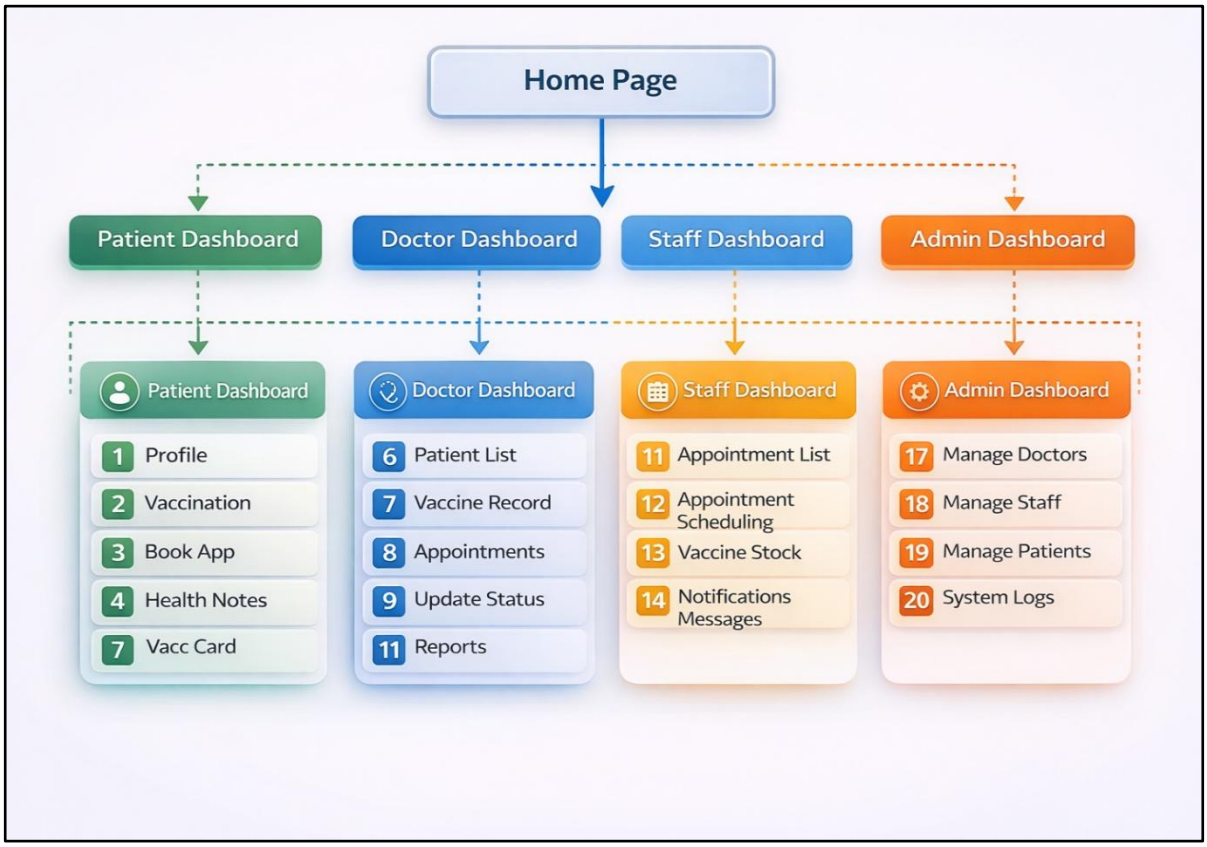


Fig 5.2 System Architecture Diagram

The System Architecture Diagram represents the overall structure of the **Baby Vaccination Management System**. It shows how different users interact with the system through their respective dashboards and how the system is organized in a structured manner.

The system starts with a **Landing Page**, from where users are directed to their respective dashboards based on their roles. The system is divided into four main modules: Patient Dashboard, Doctor Dashboard, Staff Dashboard, and Admin Dashboard. Each dashboard provides specific functionalities according to the role of the user.

The architecture ensures proper separation of functionalities, secure data handling, and smooth interaction between users and the system.

Modules in System Architecture

1. Patient Dashboard

- Manage profile details
- View vaccination schedule
- Book appointment
- View vaccination card
- Access health notes

2. Doctor Dashboard

- View patient list
- Manage vaccine records
- Handle appointments
- Update vaccination status
- Generate reports

3. Staff Dashboard

- Manage appointment scheduling
- Maintain vaccine stock
- Assist in vaccination process

4. Admin Dashboard

- Manage doctors and staff
- Manage patients
- Monitor system activities
- Maintain system logs
- Generate reports

Key Features of Architecture

- Role-based access control system
- Centralized database management
- Secure and structured data flow
- Easy navigation through dashboards
- Efficient communication between users

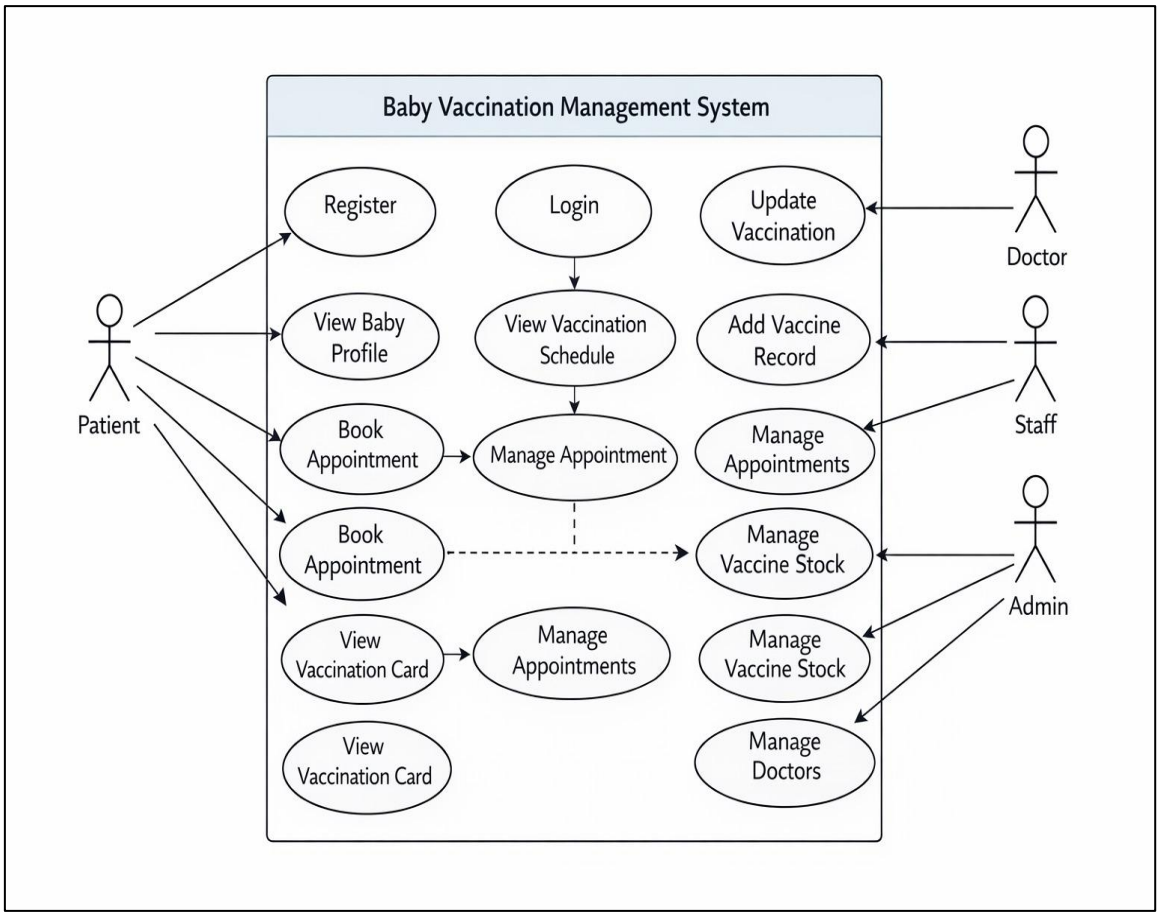


Fig. 5.3 Use Case Diagram

Actors:

- **Patient:** Represents the user (parent) who manages baby vaccination details.
- **Doctor:** Responsible for updating vaccination records and monitoring patient health.
- **Staff:** Assists in managing appointments, vaccine stock, and notifications.
- **Admin:** Manages the entire system including users, data, and reports.

Use Cases:

Authentication

- **Register:** The patient can register an account in the system.
- **Login:** The user can log in to access the system.
- **Verification:** The system verifies user credentials for secure access.

Patient Activities

- **View Baby Profile:** Patient can view baby details.
- **View Vaccination Schedule:** Patient can check upcoming vaccinations.
- **Book Appointment:** Patient can schedule vaccination appointment.
- **View Vaccination Card:** Patient can see vaccination history.
- **Receive Notifications:** Patient gets reminders for vaccines.

Doctor Activities

- **View Patient Details:** Doctor can access patient records.
- **Update Vaccination:** Doctor updates vaccination status.
- **Add Vaccine Record:** Doctor adds vaccination details.
- **Provide Health Notes:** Doctor gives medical advice.

Staff Activities

- **Manage Appointments:** Staff handles booking requests.
- **Schedule Vaccination:** Staff assigns vaccination dates.
- **Manage Vaccine Stock:** Staff maintains vaccine availability.

Admin Activities

- **Manage Patients:** Admin controls patient data.
- **Manage Doctors:** Admin manages doctor accounts.
- **Manage Staff:** Admin handles staff records.
- **Monitor System:** Admin oversees system activities.
- **Generate Reports:** Admin creates system reports.

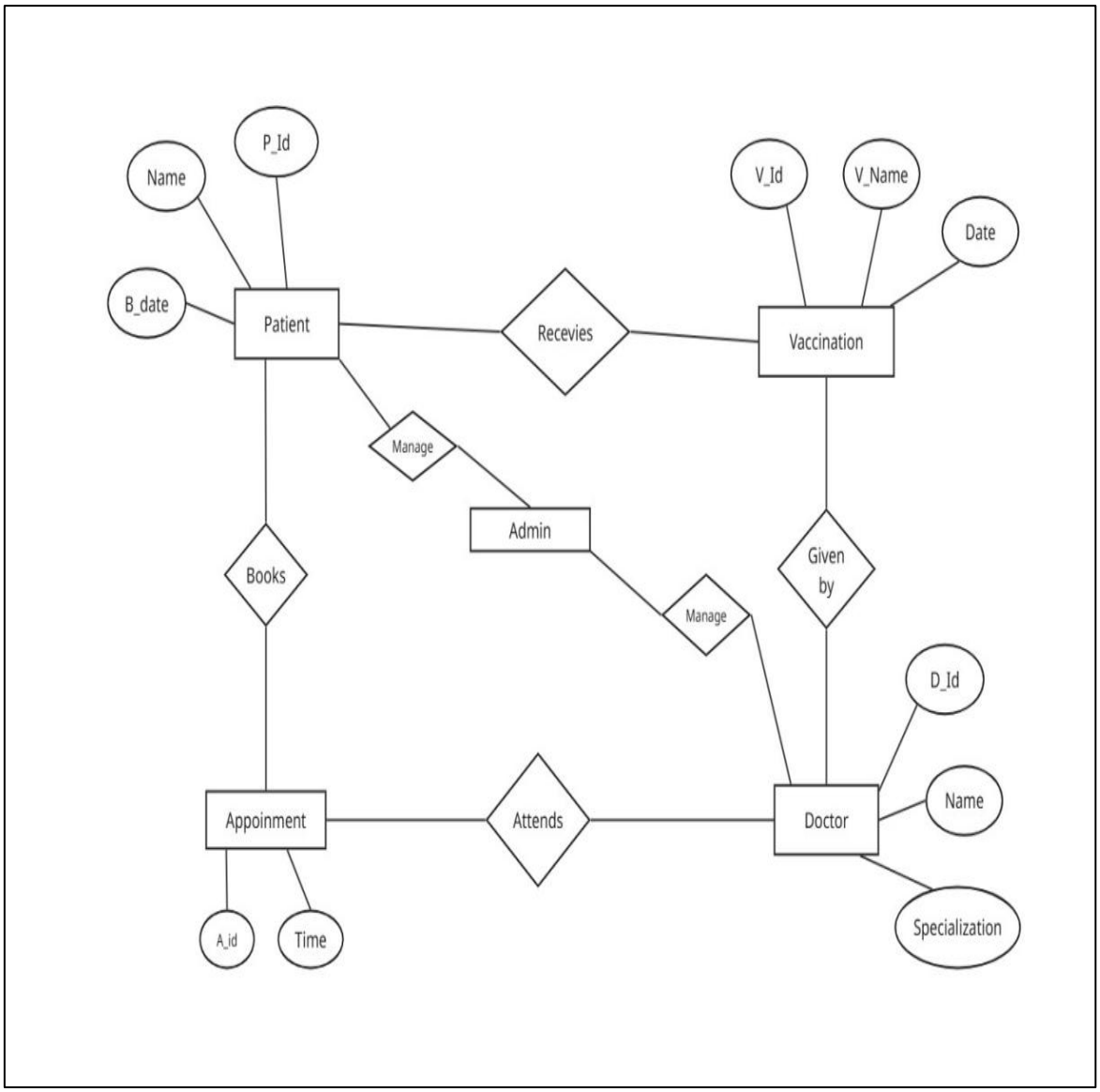


Fig. 5.4 ER Diagram

The ER diagram represents the structure of the Baby Vaccination Management System and shows the relationships between different entities such as Patient, Baby, Doctor, Staff, and Vaccination. Each patient can have multiple babies, and each baby can have multiple vaccination records. The system ensures proper data organization, efficient record management, and easy tracking of vaccination details.

CHAPTER 6

RESULTS AND APPLICATIONS

6.1 Results and Applications

The **Baby Vaccination Management System** has been successfully developed and implemented as a web-based application to manage vaccination records efficiently. The system provides a digital platform for patients, doctors, staff, and administrators to interact and manage vaccination-related activities in a structured and organized manner.

The application is developed using **HTML, CSS, JavaScript, React and Node.js**, with a backend database managed using **MongoDB**. The system ensures easy accessibility, secure data handling, and efficient management of vaccination records.

1. Patient Module Results

The patient module allows parents to manage baby vaccination details effectively.

- Patients can register and login into the system
- They can add and manage baby details
- View vaccination schedule and history
- Book vaccination appointments
- Receive reminders and notifications

This module helps parents to avoid missing vaccination dates and ensures proper healthcare tracking for babies.

2. Doctor Module Results

The doctor module improves medical record handling and efficiency.

- Doctors can view patient and baby details
- Update vaccination status after each visit
- Add vaccine details and medical notes
- Monitor vaccination progress

This reduces manual paperwork and improves accuracy in medical records.

3. Staff Module Results

The staff module helps in managing operational tasks efficiently.

- Manage appointment scheduling
- Handle patient requests
- Maintain vaccine stock details

This ensures smooth workflow and coordination between different users.

4. Admin Module Results

The admin module provides complete control over the system.

- Manage patients, doctors, and staff
- Monitor system activities
- Maintain database records
- Generate reports

Admin ensures the proper functioning and security of the system.

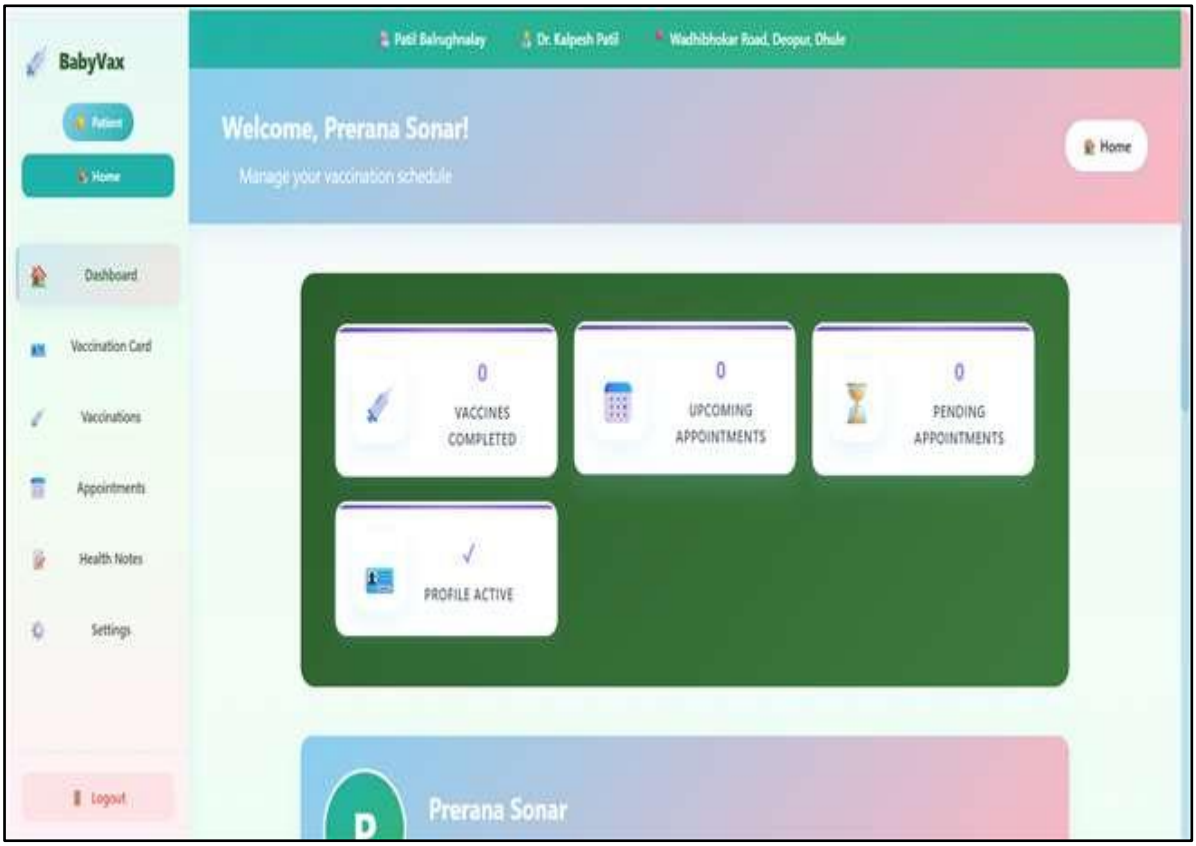


Fig 6.1: Patient Dashboard Interface

The above Diagram represents the Patient Dashboard of the Baby Vaccination Management System. It provides a user-friendly interface for parents to manage vaccination-related activities efficiently. The dashboard displays important information such as the number of vaccines completed, upcoming appointments, and pending appointments.

On the left side, a navigation menu is available which includes options like Dashboard, Vaccination Card, Vaccinations, Appointments, Health Notes, and Settings. These options help the user to easily access different features of the system.

The main section shows a summary of vaccination status through cards, allowing the patient to quickly understand the current progress. The dashboard also indicates whether the user profile is active.

Overall, this interface helps patients to track vaccination schedules, manage appointments, and stay updated with important notifications, ensuring proper healthcare management for babies

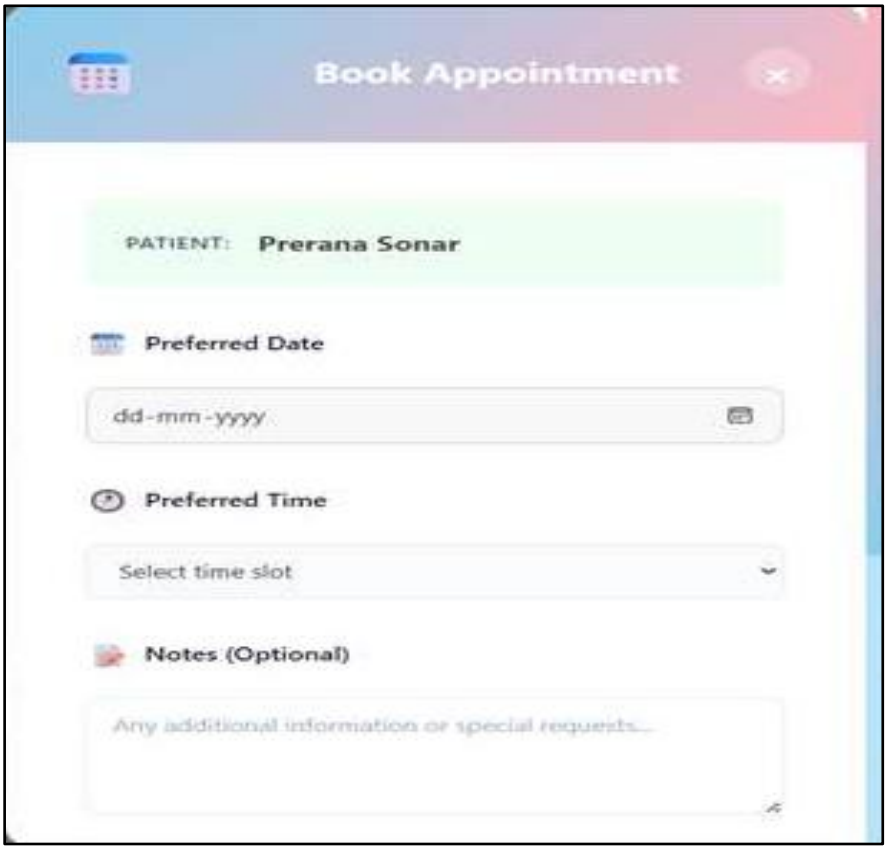


Fig 6.2: Book Appointment Interface

The above diagram represents the Book Appointment interface of the Baby Vaccination Management System. This interface allows patients to schedule vaccination appointments easily by selecting their preferred date and time. It displays the patient's name for confirmation and provides input fields such as date picker, time slot selection, and an optional notes section for additional information. The interface is designed to be simple and user-friendly, ensuring accurate data entry and smooth communication between patients and healthcare staff. This feature helps in efficient appointment scheduling and proper vaccination management.

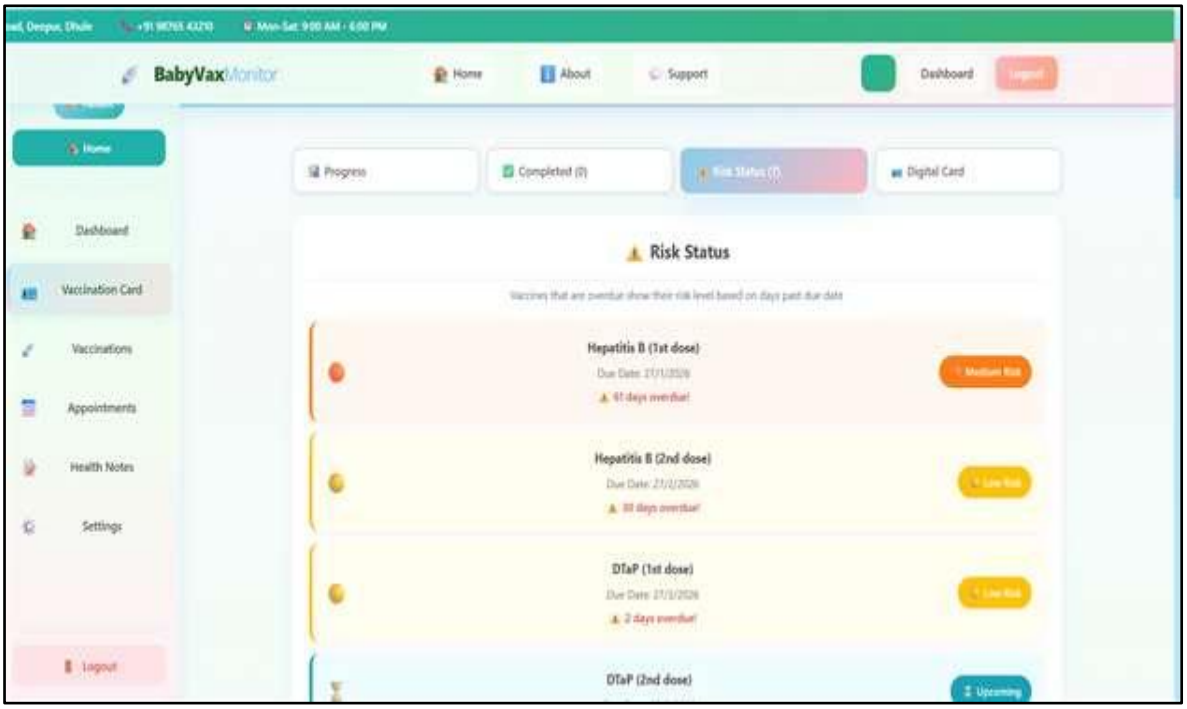


Fig 6.3: Risk Status and Vaccination Monitoring Interface

The above diagram represents the Risk Status and Vaccination Monitoring interface of the Baby Vaccination Management System. This interface provides a clear overview of vaccination progress, completed vaccines, and risk levels associated with delayed vaccinations. It highlights overdue vaccines along with the number of days delayed and categorizes them into different risk levels such as low, medium, or high risk. The system helps users to easily identify pending vaccinations and take timely action. Additionally, the interface includes navigation options for accessing vaccination cards, appointments, and other features, making it easy for patients to monitor and manage vaccination status effectively.

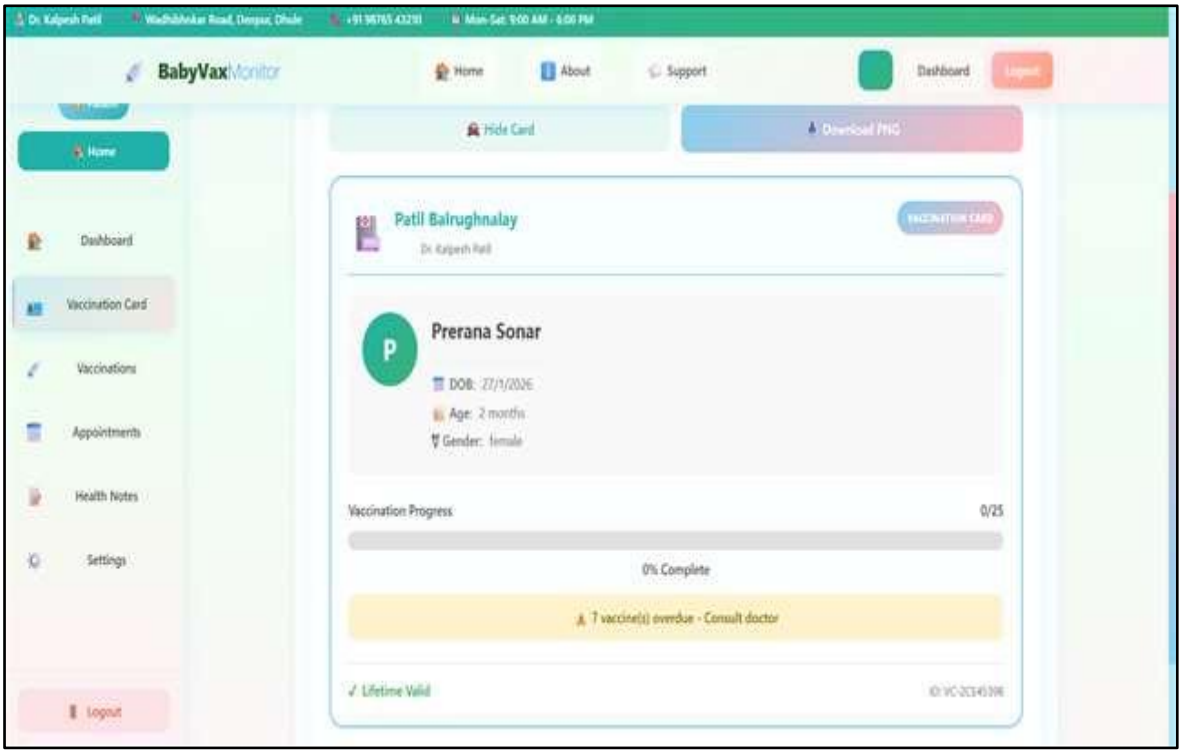


Fig 6.4: Vaccination Card Interface

The above diagram represents the Vaccination Card Interface of the Baby Vaccination Management System. This interface displays detailed information about the baby, including name, date of birth, age, and gender. It also shows the vaccination progress, indicating the number of completed vaccinations and pending doses. The system highlights overdue vaccines and provides alerts to consult a doctor if necessary. Additionally, the interface allows users to download the vaccination card and view it digitally. This feature helps patients to maintain proper vaccination records and easily track the immunization status of the baby.

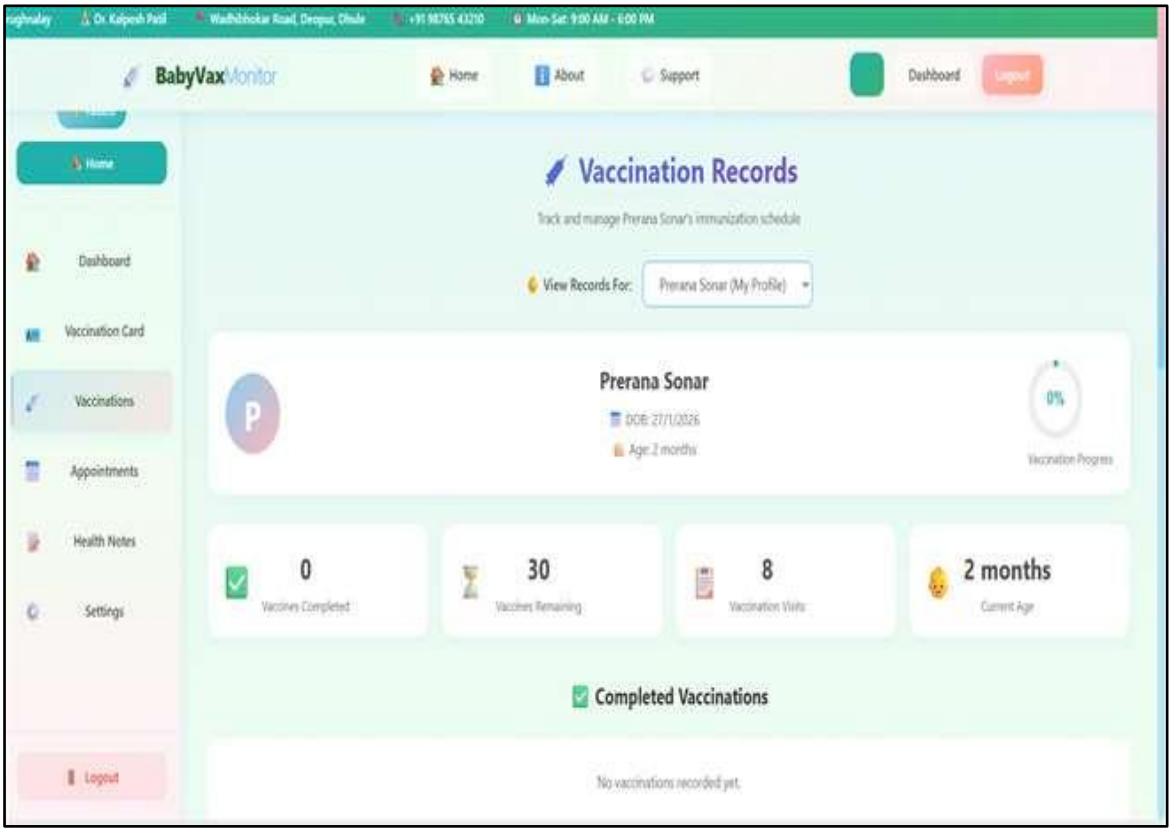


Fig 6.5: Vaccination Records Interface

The above diagram represents the Vaccination Records Interface of the Baby Vaccination Management System. This interface allows users to view and manage detailed vaccination records of the baby in an organized manner. It displays important information such as baby details, vaccination progress, number of completed vaccines, remaining vaccines, and total vaccination visits. The interface also provides a section for completed vaccinations, helping users to track immunization history effectively. This feature ensures proper monitoring of vaccination schedules and helps parents stay informed about pending and completed vaccinations, improving overall healthcare management.

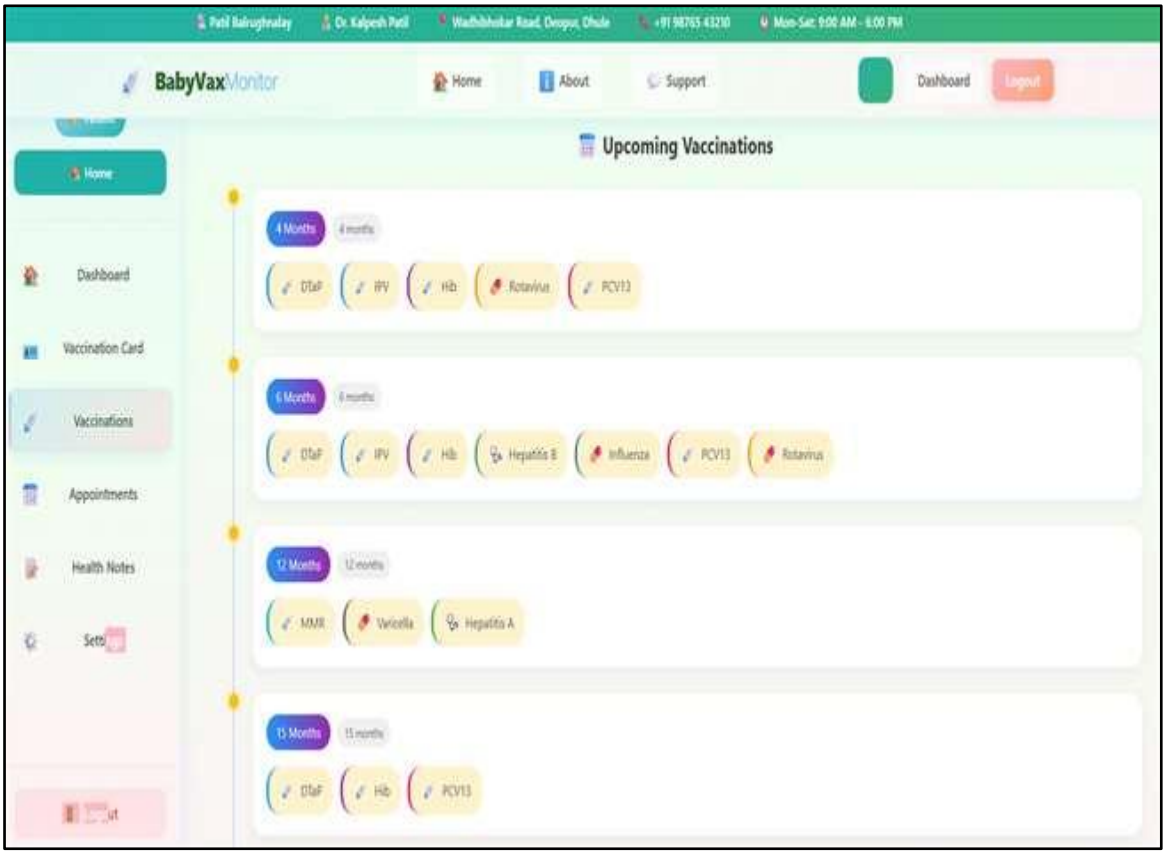


Fig 6.6: Upcoming Vaccinations Interface

The above figure represents the Upcoming Vaccinations Interface of the Baby Vaccination Management System. This interface provides a structured timeline view of upcoming vaccines based on the baby's age in months. It displays different vaccination stages such as 4 months, 6 months, 12 months, and 15 months, along with the list of required vaccines for each stage. The interface helps users easily understand which vaccines are scheduled next and plan accordingly. It improves vaccination tracking by organizing data in a clear and visually understandable format, ensuring that parents do not miss any important vaccinations.

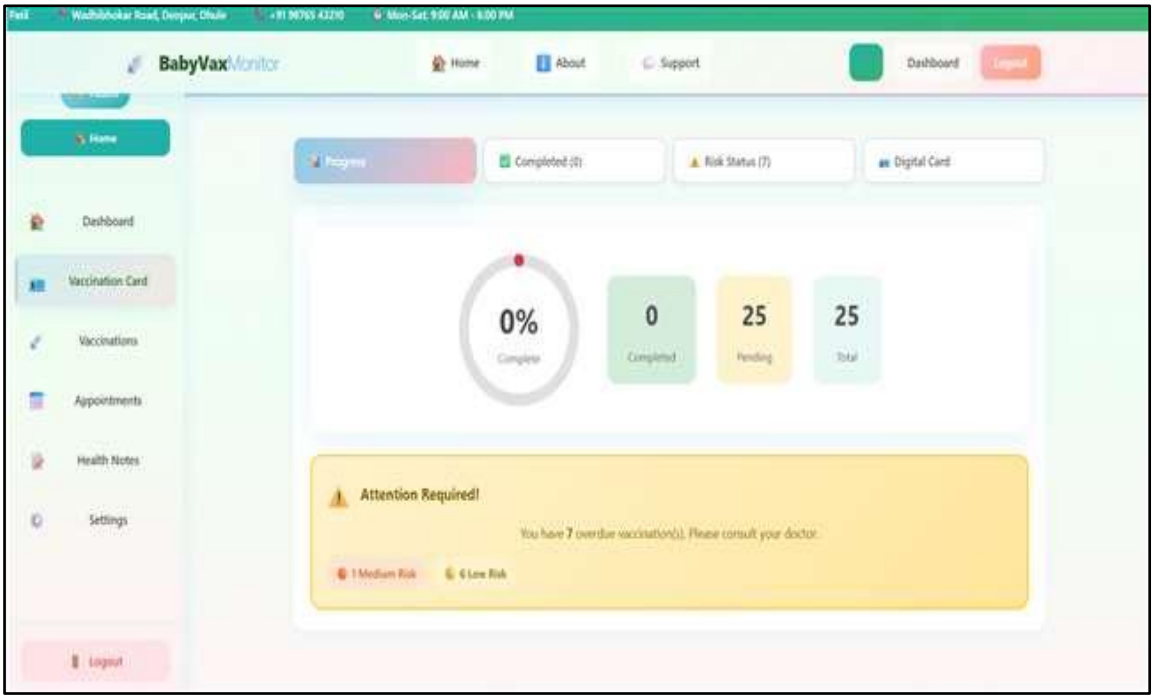


Fig 6.7: Vaccination Progress Interface

The above figure represents the Vaccination Dashboard Interface of the Baby Vaccination Management System. It provides a clear summary of vaccination progress, showing the number of completed, pending, and total vaccinations along with a visual progress indicator. The interface also highlights important alerts such as overdue vaccinations and displays risk levels like low and medium risk to indicate urgency. A side navigation panel is available for accessing different modules such as dashboard, vaccination card, appointments, and health notes. This dashboard helps users to quickly understand the vaccination status and take necessary actions, ensuring timely and effective healthcare management.

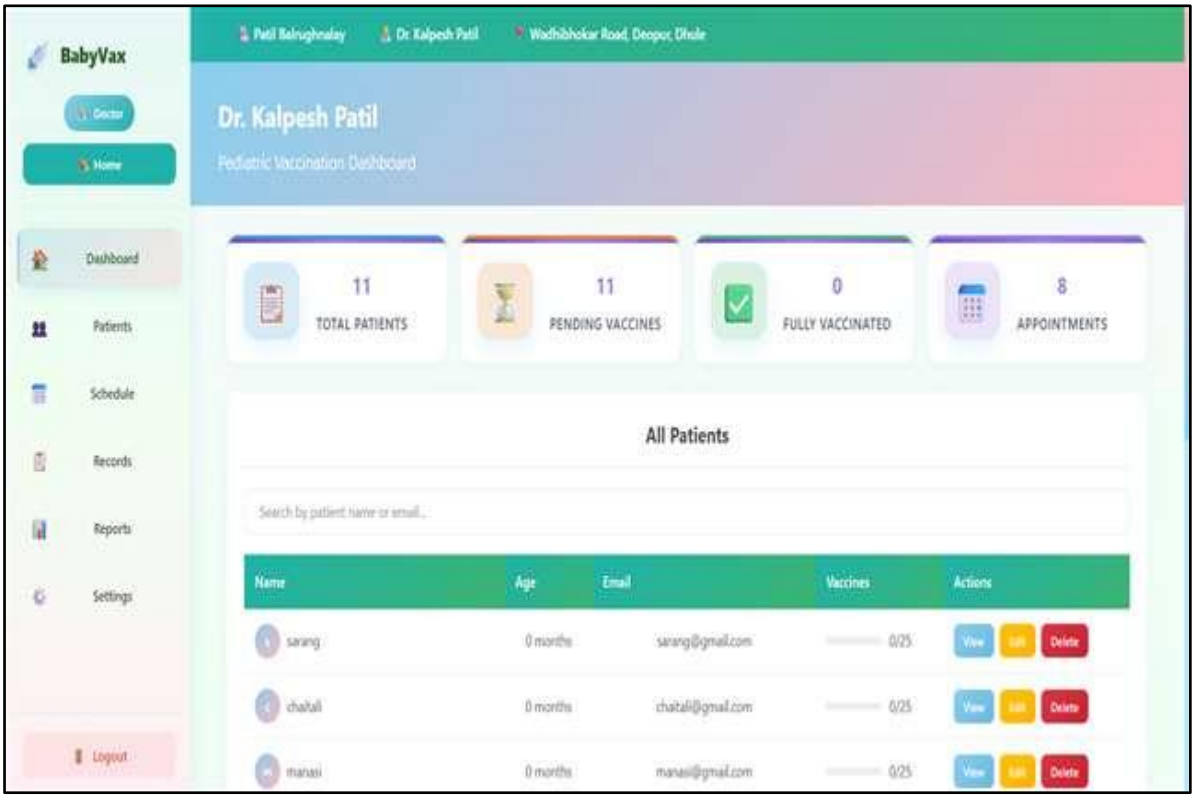


Fig 6.8: Doctor Dashboard Interface

The above figure represents the Doctor Dashboard Interface of the Baby Vaccination Management System. It provides a comprehensive overview of patient and vaccination-related data, including total patients, pending vaccines, fully vaccinated cases, and scheduled appointments. The dashboard also displays a list of patients with details such as name, age, email, and vaccination progress, along with action options like view, edit, and delete. A side navigation panel allows the doctor to access modules such as patients, schedule, records, and reports. This interface helps doctors efficiently manage patient records, monitor vaccination status, and streamline healthcare operations.

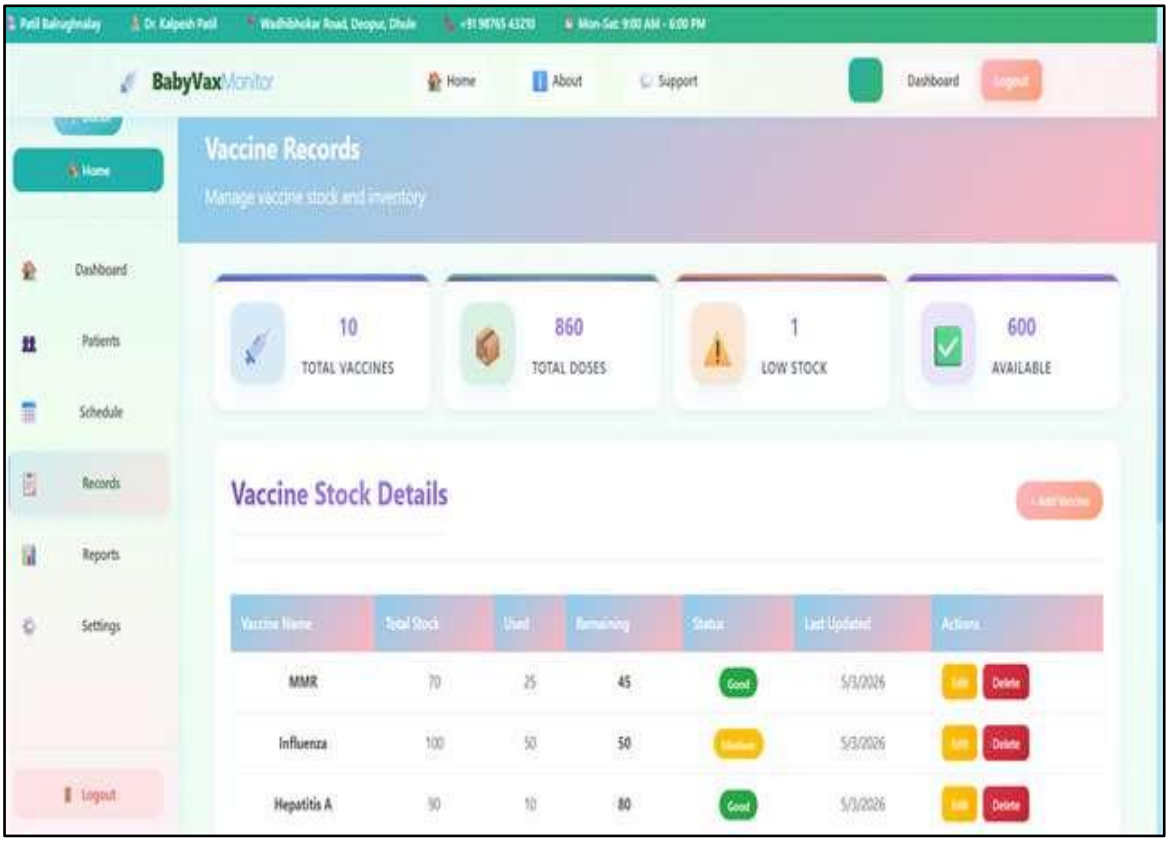


Fig 6.9: Vaccine Records and Stock Management Interface

The above figure represents the Vaccine Records and Stock Management Interface of the Baby Vaccination Management System. This interface provides detailed information about vaccine inventory, including total vaccines, total doses, available stock, and low stock alerts. It also displays a structured table containing vaccine details such as vaccine name, total stock, used doses, remaining quantity, status, and last updated date. Action buttons like edit and delete are provided for managing records efficiently. The dashboard helps healthcare staff and administrators to monitor vaccine availability, manage stock levels, and ensure timely replenishment, thereby supporting smooth vaccination operations.

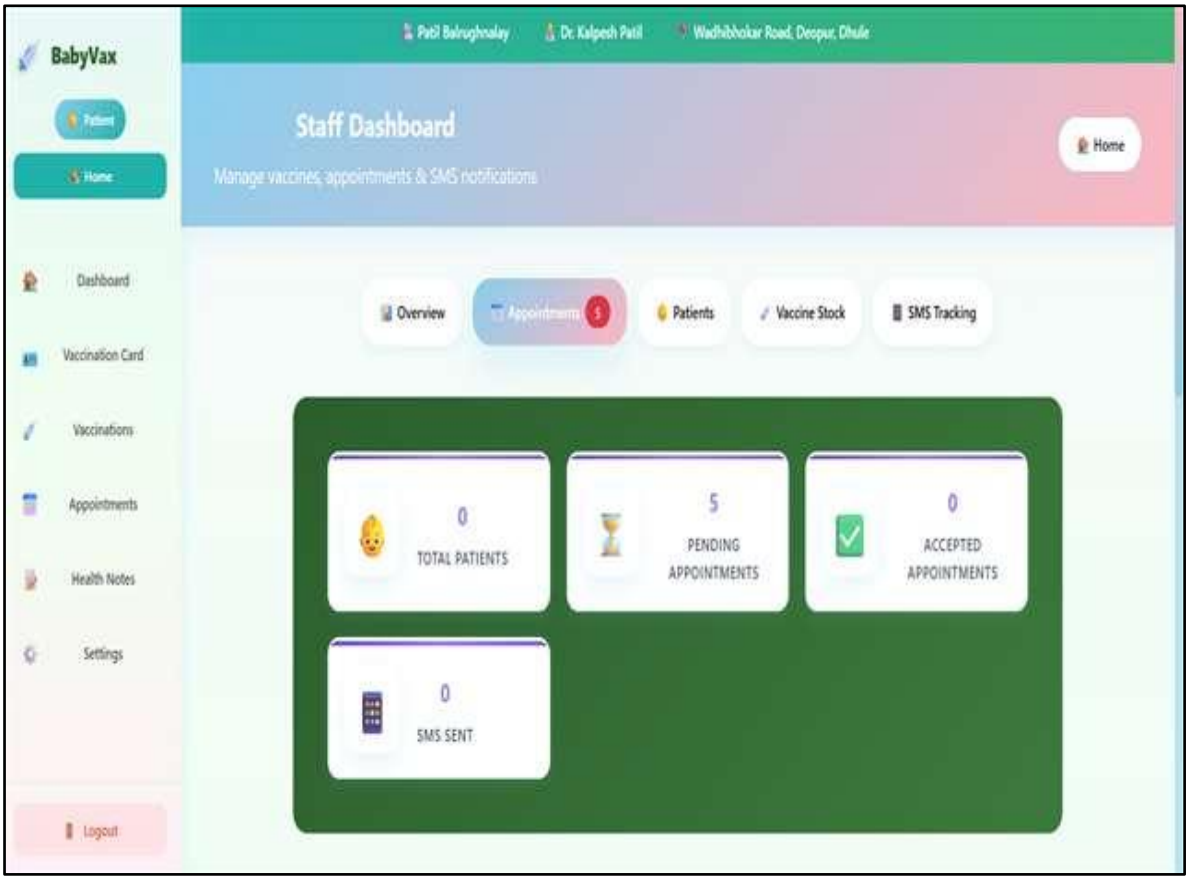


Fig 6.10: Staff Dashboard Interface

The above figure represents the Staff Dashboard Interface of the Baby Vaccination Management System. This interface is designed to help staff efficiently manage vaccination-related activities such as appointments, patient records, vaccine stock, and SMS notifications. It displays key information including total patients, pending appointments, accepted appointments, and SMS notifications sent. The dashboard also provides quick navigation options like overview, appointments, patients, vaccine stock, and SMS tracking. A side menu is available for accessing different modules such as dashboard, vaccination card, appointments, and health notes. This interface helps staff to streamline operations, manage schedules effectively, and ensure smooth communication with patients.

Fig 6.11: Appointment Management Interface

The above figure represents the Appointment Management Interface of the Baby Vaccination Management System. This interface displays a list of scheduled appointments along with details such as date, time, and patient name. Each appointment is marked with its current status, such as pending, and provides action buttons like accept and reject for quick decision-making. The interface is designed to help staff or doctors efficiently manage daily appointments and respond to requests in real time. It improves scheduling efficiency, reduces delays, and ensures proper coordination between patients and healthcare providers.

CHAPTER 7

TESTING AND DEFECT REPORT

1. Introduction

The Test Plan for the Baby Vaccination Management System outlines the overall strategy, scope, objectives, and methodology for testing the system. Testing is a crucial phase in the software development lifecycle, as it ensures that the system functions correctly, meets user requirements, and is free from critical defects. This test plan provides a structured approach to verify and validate the system functionalities.

2. Objectives of Testing

The main objectives of testing the system are as follows:

- To verify that all modules of the system are working as expected
- To identify defects, errors, and inconsistencies in the system
- To ensure that the system meets functional and user requirements
- To validate data accuracy and system performance
- To improve system reliability, usability, and efficiency

3. Scope of Testing

The testing process covers all major modules of the Baby Vaccination Management System, including:

- User Registration and Login Module
- Baby Profile Management Module
- Appointment Booking and Scheduling Module
- Vaccination Records Management Module
- Dashboard and Reporting Module

4. Testing Methodology / Approach

The following testing approach was used:

- Manual Testing: All functionalities were tested manually using predefined test cases
- Functional Testing: Each feature was tested to ensure correct operation
- Validation Testing: Input fields were tested with valid and invalid data

• Integration Testing: Interaction between modules was verified

• System Testing: Complete system testing was performed to ensure end-to-end functionality

This approach helped in identifying defects at different levels of the system.

5. Test Environment

The testing was performed in the following environment:

• Operating System: Windows 10/11

• Frontend Technologies: HTML, CSS, JavaScript, React

• Backend Technology: Node.js

• Database: MongoDB

This environment ensured proper execution and validation of all system functionalities.

6. Test Data

Various types of test data were used to validate system behaviour:

• Valid login credentials

• Invalid username and password combinations

• Sample baby details (name, DOB, parent details)

• Appointment data (date, time, notes)

• Vaccination records (vaccine name, date, status)

• Edge cases such as empty fields, incorrect inputs, and duplicate entries

7. Entry Criteria

Testing was initiated based on the following conditions:

• Development of all modules was completed

• Database was properly configured

• Required test cases were prepared

• System was deployed in a testing environment

8. Exit Criteria

Testing was considered complete when:

- All major test cases were executed
- Critical and high severity defects were resolved
- System performance was stable
- The application met all functional requirements

9. Risks and Challenges

During testing, the following risks and challenges were identified:

- Possibility of data validation errors
- Performance issues due to large data
- UI inconsistencies on different devices
- Dependency on internet and server availability

10. Limitations of Testing

- Testing was conducted in a controlled environment
- Real-time user load testing was limited
- Some minor UI issues may still exist
- External integrations were not fully tested

11. Conclusion

The Test Plan provided a systematic and structured approach for testing the Baby Vaccination Management System. It helped in identifying defects, improving system performance, and ensuring that all functionalities work as expected. After successful testing and fixing of identified issues, the system was found to be reliable, efficient, and ready for deployment.

Testing and Validation

Testing is a crucial phase in the development of the Baby Vaccination Management System. It ensures that the system functions correctly, meets user requirements, and is free from errors. Various types of testing were conducted to verify the performance, functionality,

and reliability of the system. The main objective of testing is to identify defects and ensure smooth operation of the application.

- **Types of Testing Performed**

1) Unit Testing – Individual modules like login, registration, and appointment booking were tested separately
2) Integration Testing – Interaction between modules such as patient, doctor, and staff was tested
3) System Testing – Complete system functionality was tested
4) User Acceptance Testing – System was tested by users for real-time validation

Table 1 Test cases

Test Case ID	Module	Description	Input	Expected Output	Actual Output	Status
TC01	Login	Valid login credentials	username & password	Login successful	Login successful	Pass
TC02	Login	Invalid password	wrong password	Error message	Error displayed	Pass
TC03	Registration	Empty fields	blank input	Validation error	Error shown	Pass
TC04	Baby Module	Add baby details	valid data	Data saved	Data saved	Pass
TC05	Baby Module	Missing DOB	empty DOB	Error message	Error shown	Pass
TC06	Appointment	Book appointment	valid date/time	Appointment booked	Booked	Pass

Test Case ID	Module	Description	Input	Expected Output	Actual Output	Status
TC07	Appointment	Past date	old date	Not allowed	Error shown	Pass
TC08	Vaccination	Update record	valid data	Updated	Updated	Pass
TC09	Vaccination	Missing vaccine name	empty field	Error	Accepted	Fail
TC10	Dashboard	Load dashboard	user login	Data displayed	Loaded	Pass
TC11	Notifications	Reminder alert	scheduled vaccine	Notification sent	Sent	Pass
TC12	Reports	Generate report	date range	Report generated	Generated	Pass
TC13	Profile	Update profile	valid input	Updated	Updated	Pass
TC14	Login	SQL injection test	malicious input	Blocked	Blocked	Pass
TC15	Appointment	Double booking	same slot	Not allowed	Allowed	Fail

Failed Test Cases and Defects

During the testing phase, several defects were identified in the system which affected functionality and user experience. These issues were analysed carefully to improve system performance and reliability.

The vaccination module allowed submission of records with empty or incomplete fields in certain scenarios, indicating insufficient input validation.
The appointment system permitted double booking for the same date and time slot, which could lead to scheduling conflicts and operational issues.

There was a minor delay observed while loading the dashboard, especially when handling larger data sets, affecting system responsiveness.

These defects were further reviewed and necessary corrective measures were taken to enhance the overall system performance and user experience.

Table 2 Defect Report

Defect ID	Module	Description	Severity	Status
D01	Vaccination	System allowed submission of vaccination records with empty or missing fields due to improper validation	Medium	Fixed
D02	Appointment	System allowed multiple bookings for the same date and time slot, causing scheduling conflicts	High	Fixed
D03	Dashboard	Dashboard experienced slight delay while loading data, affecting user experience	Low	Fixed
D04	UI	User interface alignment issues observed on different screen sizes, especially in mobile view	Low	Pending

Testing Summary

The Baby Vaccination Management System was thoroughly tested using multiple test cases that covered all major functionalities of the application, including login, registration, baby management, appointment booking, vaccination updates, and report generation. The primary objective of testing was to ensure that the system operates correctly, efficiently, and according to user requirements.

During the testing process, most of the test cases were executed successfully, indicating that the core functionalities of the system are working as expected. The system was able to handle user inputs properly, perform required operations, and display accurate results in different modules.

However, a few defects were identified during testing. These issues included validation errors, appointment scheduling conflicts, minor performance delays, and user interface inconsistencies. Each defect was carefully analysed based on its severity and impact on the system. Appropriate corrective measures were taken to resolve critical and medium-level issues, while minor issues were noted for further improvement.

After implementing the necessary fixes, the system showed significant improvement in terms of performance, reliability, and usability. The final testing results indicate that the system is stable, user-friendly, and capable of handling real-time operations effectively.

Overall, the testing phase played a crucial role in enhancing the quality of the system and ensuring that it meets the desired functional and performance standards.

CHAPTER 8

CONCLUSION AND FUTURE SCOPE

Conclusion

In conclusion, the **Baby Vaccination Management System** provides an efficient and user-friendly platform for managing vaccination records and healthcare activities. The system helps patients (parents), doctors, staff, and administrators to interact in a structured manner and perform their respective tasks effectively.

By digitizing vaccination records, the system reduces manual work, minimizes errors, and ensures accurate data management. It allows parents to track vaccination schedules, receive timely reminders, and maintain proper health records of their babies. Doctors and staff can efficiently manage appointments, update vaccination details, and monitor patient progress.

Overall, the system improves healthcare management, ensures timely vaccination, and provides a reliable solution for maintaining vaccination data in an organized way.

Future Scope

1. The system can be expanded for use across hospitals and healthcare centre's all over India
2. Multi-language support can be added to make the system accessible in different regional languages
3. Mobile application (Android/iOS) can be developed for better accessibility
4. SMS and email notification system can be integrated for reminders
5. Integration with government health systems for centralized data management
6. Cloud-based storage can be implemented for better data security and scalability
7. AI-based prediction system can be added for health monitoring and risk analysis

CHAPTER 9

REFERENCES AND BIBLIOGRAPHY

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